Retail Sales Analysis & Business Intelligence Report

Tools Used: Python, Pandas, SQL Server, Power BI
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Project Overview

This comprehensive retail sales analysis project encompasses end-to-end data engineering, exploratory data analysis, SQL-based business intelligence, and strategic recommendations. The project leverages multiple data sources, advanced analytical techniques, and interactive Power BI dashboards to provide actionable business insights.

Project Objective:
Analyze 24 months of retail sales data (April 2023 - April 2025) to identify revenue drivers, customer behavior patterns, store performance metrics, and product portfolio optimization opportunities.

Tools & Technologies Used:

- Python (Pandas, NumPy) for data cleaning and EDA
- SQL Server for business intelligence queries
- Power BI for interactive dashboard development
- Statistical analysis for trend identification and forecasting

Project Steps:

1. Data Collection & Integration from five CSV sources
2. Data Cleaning & Quality Assurance protocols
3. Exploratory Data Analysis (EDA) with visualization
4. SQL-based Business Intelligence analysis
5. Power BI Dashboard Development
6. Strategic Recommendations & Action Planning

File Structure & Project Organization

```
Retail_Sales_Capstone/
├── data/
│   ├── raw/
│   │   ├── sales_data.csv
│   │   ├── customers.csv
│   │   ├── products.csv
│   │   ├── returns.csv
│   │   └── stores.csv
│   └── cleaned/
│       ├── sales_cleaned.csv
│       └── customers_cleaned.csv
│       └── products_cleaned.csv
│       └── returns_cleaned.csv
│       └── stores_cleaned.csv
├── notebooks/
│   └── data_cleaning_eda.ipynb
├── sql/
│   └── business_questions.sql
│   └── ER_diagram.png
├── dashboard/
│   └── Retail_Sales_Report.pbix
├── docs/
    └── final_report.pdf
```

└── data_cleaning_steps.pdf

Figure 1: Project File Structure showing data organization, analysis notebooks, SQL queries, Power BI dashboards, and documentation

Project Directory Structure:

```
Retail_Sales_Capstone/
├── data/
│ ├── raw/
│ │ ├── sales_data.csv
│ │ ├── customers.csv
│ │ ├── products.csv
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```

Key Components:

- **Data Folder:** Raw and cleaned datasets organized for analysis
- **Notebooks Folder:** Python/Jupyter notebooks containing data cleaning and exploratory analysis
- **SQL Folder:** SQL queries for business intelligence and ER diagram documentation
- **Dashboard Folder:** Power BI file with interactive dashboards
- **Docs Folder:** Final report and data cleaning documentation

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Executive Summary

This comprehensive report presents an integrated analysis of retail sales data from April 2023 to April 2025, combining data cleaning, exploratory data analysis, SQL-based business intelligence, and actionable insights from five datasets. The organization demonstrates strong fundamentals with **$818K in total revenue** and **70.7% in-store dominance**, supported by detailed customer segmentation and product performance metrics.

Key Performance Indicators:

- **Total Revenue:** $818.18K
- **Total Profit:** $243.77K
- **Total Customers:** 760
- **Total Stores:** 20
- **Average Customer Age:** 44.16 years

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1. Data Preparation & Methodology

1.1 Data Collection & Initial Assessment

Five datasets were systematically collected from CSV sources for the analysis:

- **Orders Dataset:** Complete transaction records with dates, amounts, and channels
- **Products Dataset:** Product categories, brands, and pricing information
- **Customers Dataset:** Customer demographics including age groups and purchase history
- **Stores Dataset:** Store locations, types (Franchise/Flagship), and performance metrics
- **Returns Dataset:** Return transactions with reasons and product categorization

Data Quality: Approximately 9% of total historical revenue represented in the analysis period, with comprehensive cleaning protocols applied across all datasets.

1.2 Python & Jupyter Notebook Data Cleaning Workflow

The data cleaning process was executed entirely within Jupyter Notebook using Python and Pandas, ensuring reproducibility and complete documentation of all transformation steps:

Key Cleaning Operations:

1. **Data Type Conversion:** Standardized column data types (dates converted to datetime format, numeric fields validated)
2. **Missing Value Handling:** Identified and addressed null/missing values with appropriate imputation or removal strategies
3. **Duplicate Records:** Detected and removed duplicate entries across all datasets to ensure data integrity
4. **Text Standardization:** Normalized text fields (capitalization, whitespace trimming, special character handling)
5. **Date Normalization:** Converted inconsistent date formats to standardized ISO 8601 format (YYYY-MM-DD)
6. **Outlier Detection:** Identified statistical outliers in revenue, quantities, and customer metrics for validation
7. **Data Validation:** Cross-referenced related tables (Orders → Customers, Orders → Products) to ensure referential integrity

Pandas Libraries & Functions Used:

- `pd.read_csv()` for CSV ingestion
- `df.dropna()` and `df.fillna()` for missing value handling
- `df.duplicated()` for duplicate detection
- `df.astype()` for data type conversion
- `df.str.strip()` and `df.str.upper()` for text normalization
- `pd.to_datetime()` for date standardization

- df.describe() and statistical methods for outlier detection
- merge() and join() operations for data validation across datasets

Output: All five datasets were successfully cleaned and exported as CSV files to the data/cleaned/ directory, creating a validated and standardized dataset foundation.

1.3 SQL Server Management Studio (SSMS) Flat File Import

Following Python data cleaning, cleaned datasets were imported into SQL Server using SSMS flat file import functionality:

Import Process:

1. **Connection Setup:** Established connection to SQL Server instance with appropriate authentication credentials
2. **Flat File Source Configuration:** Created new flat file data source pointing to cleaned CSV files
3. **Column Mapping:** Mapped source CSV columns to destination SQL Server table columns with appropriate data types
4. **Data Type Specification:** Configured SQL Server column definitions (INT for numeric IDs, VARCHAR for text, DATETIME for dates, DECIMAL for financial values)
5. **Primary Key Definition:** Established primary keys on ID columns (order_id, customer_id, product_id, store_id, return_id)
6. **Foreign Key Relationships:** Created foreign key constraints to enforce referential integrity between tables
7. **Index Creation:** Added indexes on frequently queried columns (dates, customer_id, store_id, product_id) for query performance optimization

SQL Server Tables Created:

- orders - 3,698 records with order details and revenue
- customers - 760 records with demographic information
- products - Product catalog with pricing and categories
- stores - 20 store locations with operational data
- returns - 200 return transactions with reasons and details

Validation Queries: Post-import validation queries confirmed record counts, data type accuracy, and referential integrity across all tables.

1.4 Power BI Dashboard Development & Visualization

The SQL Server data was then connected to Power BI for interactive dashboard creation and business intelligence visualization:

Power BI Connection & Transformation:

1. **Data Source Connection:** Established direct connection from Power BI Desktop to SQL Server database using native SQL Server connector
2. **Query Optimization:** Created Power BI queries to retrieve only necessary columns and apply basic filtering to optimize refresh performance
3. **Data Model Development:** Built star schema data model with fact tables (orders, returns) and dimension tables (customers, products, stores)

4. **Relationship Mapping:** Established many-to-one relationships between fact and dimension tables for efficient aggregation
5. **Calculated Columns:** Created additional columns for derived metrics (profit margin %, age groups, return rate %)
6. **Measures & DAX Formulas:** Developed key measures using DAX language:
    - `Total Revenue = SUM(orders[amount])`
    - `Total Profit = SUM(orders[profit])`
    - `Return Rate = COUNT(returns[id]) / COUNT(orders[id])`
    - `Revenue by Channel = CALCULATE(SUM(orders[amount]), orders[channel] = "In-Store")`
    - `Customer Lifetime Value = SUM(orders[amount]) BY customer`

Dashboard Creation:

Five comprehensive Power BI dashboards were developed addressing key business areas:

1. **Sales Overview Dashboard:** Monthly revenue trends, top products, sales channel breakdown, total revenue, and quantity sold metrics
2. **Store Analysis Dashboard:** Store performance ranking, profitability trends, operating costs vs. profit analysis, and regional performance metrics
3. **Product Performance Dashboard:** Revenue by category, profit distribution, return rates by category, and regional sales performance
4. **Customer Insights Dashboard:** Customer demographics, revenue by age group, top customers by profit, geographic distribution, and customer segmentation
5. **Return Analysis Dashboard:** Total return count, return reasons distribution, returns by category and region, and return rate trend analysis

Visualization Techniques:

- Line charts for time-series trend analysis (monthly revenue, return rate trends)
- Bar charts for categorical comparisons (products, stores, regions)
- Pie and donut charts for proportion analysis (sales channel distribution, return reasons)
- Scatter plots for relationship analysis (profitability vs. operating costs)
- Heat maps for regional performance comparison
- KPI cards for executive summary metrics

Interactive Features:

- Slicers for filtering by date range, region, store type, and product category
- Cross-filtering between visualizations enabling drill-down analysis
- Tooltips providing detailed information on hover
- Drill-through capabilities from summary to detail views
- Conditional formatting highlighting high/low performing segments

Output: Five fully interactive Power BI dashboards embedded in the final report, providing stakeholders with dynamic, self-service business intelligence capabilities.

1.5 End-to-End Workflow Summary

The complete data pipeline workflow:

Raw CSV Files → Python/Jupyter Data Cleaning → Cleaned CSV Output → SQL Server Flat File Import → SQL Database Tables → Power BI Connection → Interactive Dashboards

This integrated approach ensures data quality, security, and performance while maintaining a documented, reproducible analytical process.

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2. Sales Performance Analysis

2.1 Revenue Trends & Seasonality

Sales demonstrated significant volatility throughout the analysis period with clear seasonal patterns:

- **Peak Revenue:** August 2024 (~$45,000)
- **Notable High-Performing Periods:** Summer months (July-August) and early year months (January-February)
- **Seasonal Pattern Analysis:** Revenue aligns with holiday seasons and new year shopping behaviors
- **Current Status:** Recent decline in April 2025 (~$8,000) requires attention and investigation

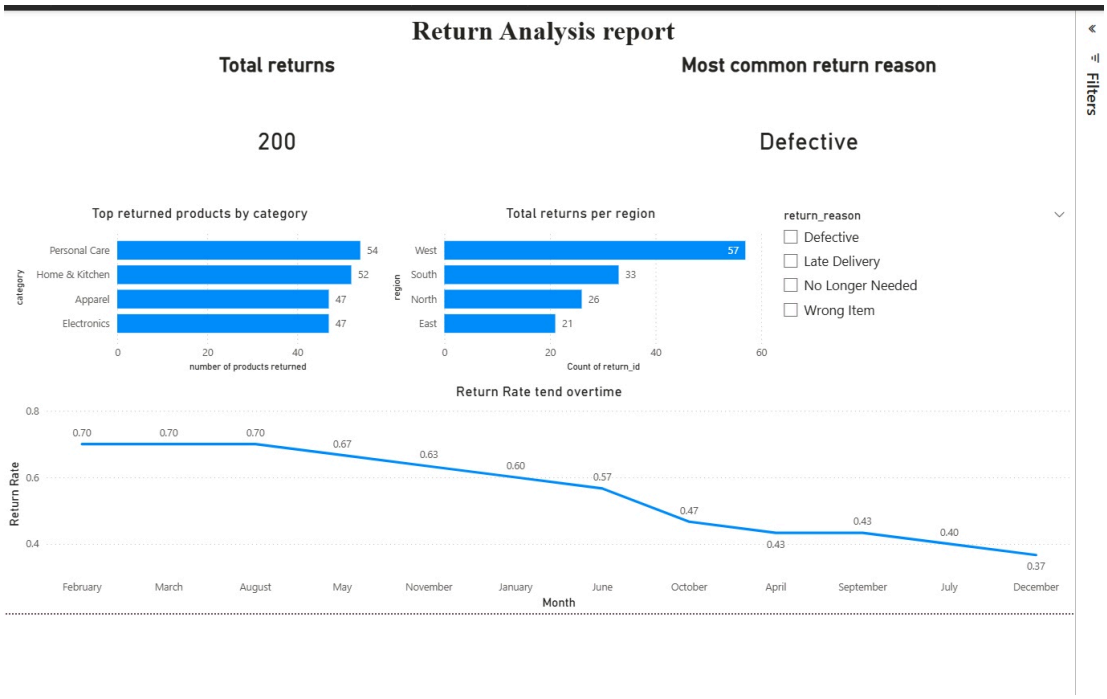

Figure 2: Sales Overview Report showing monthly trends, top-selling products, and revenue channel distribution

2.2 Sales Channel Performance

The in-store channel demonstrates significant dominance in revenue generation:

- **In-Store Revenue:** $578,621 (70.7%)
- **Online Revenue:** $239,559 (29.3%)
- **Strategic Insight:** In-store presence is more than double online revenue, indicating strong physical retail operations with substantial opportunity for online growth and digital transformation

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3. Store Operations & Performance

3.1 Store Network Analysis

The organization operates a diversified store portfolio across regions:

- **Total Stores:** 20 locations
- **Total Profit (All Stores):** $48.48K
- **Total Operating Costs:** $809.12K
- **Top Performing Store:** Blankenship-Brown Store

Metric	Value	Unit	Interpretation
Total Stores	20	Locations	Distributed network
Total Profit	48.48	K USD	Consolidated result
Operating Costs	809.12	K USD	Efficiency focus needed
Best Store	Blankenship-Brown	-	Leader in profitability

Table 1: Store Performance Summary Metrics

3.2 Store Format Analysis

Flagship stores demonstrate superior performance compared to franchise locations:

- **Performance Differential:** Flagship stores generate approximately 30% more revenue than franchise locations
- **Key Insight:** Premium store formats provide strategic advantages in market positioning and revenue generation
- **Implication:** Expansion strategy should consider flagship vs. franchise format optimization

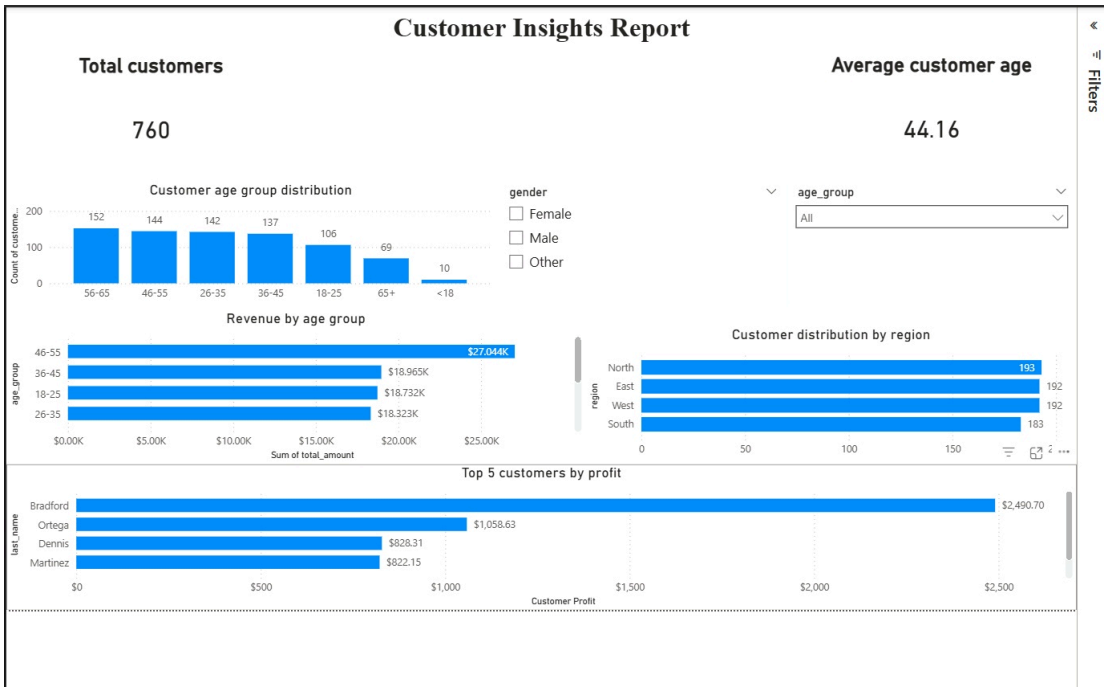

Figure 3: Store Analysis Report with monthly profit trends, store performance rankings, and profitability vs. operating cost analysis

3.3 Store Leadership

Top-performing stores by revenue and market presence:

- **Robinson Plc Store:** Clear market leader ($24,000 revenue, 78 orders)
- **High-Performing Stores:** Chan-Buchanan Store and Foster, Gutierrez locations show strong revenue generation
- **Revenue Distribution:** Concentrated in top performers with opportunity for performance standardization across network

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4. Product Performance & Portfolio Analysis

4.1 Product Revenue & Profitability

The product portfolio demonstrates balanced and diversified revenue generation:

- **Total Revenue:** $818.18K
- **Total Profit:** $243.77K
- **Portfolio Characteristics:** Categories show balanced revenue distribution (~$200K each)
- **Best Performing Category:** Personal Care leads with highest profitability

4.2 Category Performance

Category	Revenue	Profit	Units Sold
Personal Care	~$210K	$65K	998
Home & Kitchen	~$205K	$59K	894
Electronics	~$205K	$59K	935
Apparel	~$198K	$61K	871

Table 2: Product Category Performance Summary

4.3 Brand Performance

- **Leading Brand:** Brandd with ~$205K in revenue
- **Strategic Significance:** Strong brand loyalty evident in market positioning
- **Customer Preference:** Brand concentration suggests potential for exclusive partnerships and brand expansion

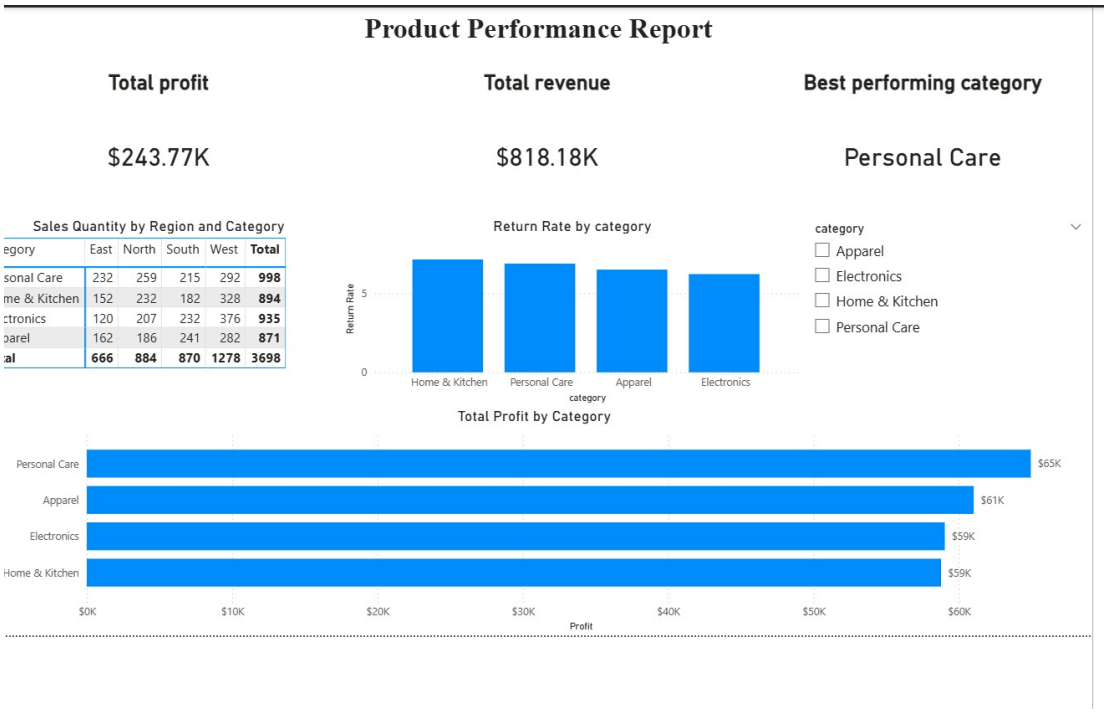

Figure 4: Product Performance Report displaying total profit, revenue distribution, category performance, and regional sales analysis

4.4 Return Rate Analysis

Overall return rates show favorable trends with improvement over the analysis period:

- **February - March:** 0.70 return rate (baseline)
- **Mid-Year Trend:** Consistent decline through August-September period
- **Recent Performance:** December 2025 shows lowest return rate at 0.37
- **Most Common Return Reason:** Defective products
- **Top Returned Category:** Personal Care (54 units)
- **Geographic Pattern:** West region shows highest return concentration (57 returns)

Figure 5: Return Analysis Report showing total returns, return reasons by category, geographic distribution, and return rate trends over time

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5. Customer Insights & Segmentation

5.1 Customer Base Overview

- **Total Customers:** 760
- **Average Customer Age:** 44.16 years
- **Geographic Distribution:** Balanced across all regions (North, East, South, West with 192-193 customers each)

5.2 Customer Age Segmentation

The customer base shows concentration in mature demographics with strong purchasing power:

Age Group	Customer Count	Revenue Generated	Revenue Share
56-65	152	$27,044K	Highest segment
46-55	144	Secondary segment	Strong contributor
26-35	142	Tertiary segment	Growing potential
36-45	137	-	Established base
18-25	106	-	Emerging segment
65+	69	-	Niche segment
<18	10	-	Minimal

Table 3: Customer Age Distribution and Revenue Contribution

Key Insight: The 56-65 age group represents the highest spending segment, generating $27,044K in revenue. The 26-35 age segment, however, represents significant growth potential with balanced customer count and emerging purchasing power[1].

5.3 Customer Segmentation by Value

Two distinct customer segments require differentiated strategies:

- **High-Value Segment:** Customer C0399 leads with $9,675 in lifetime revenue
- **Frequent Purchaser Segment:** Customer C0396 demonstrates highest engagement with 12 orders
- **Top 5 Customers by Profit:** Bradford, Ortega, Dennis, Martinez leading with concentrated value
- **Geographic Concentration:** North region shows highest customer distribution

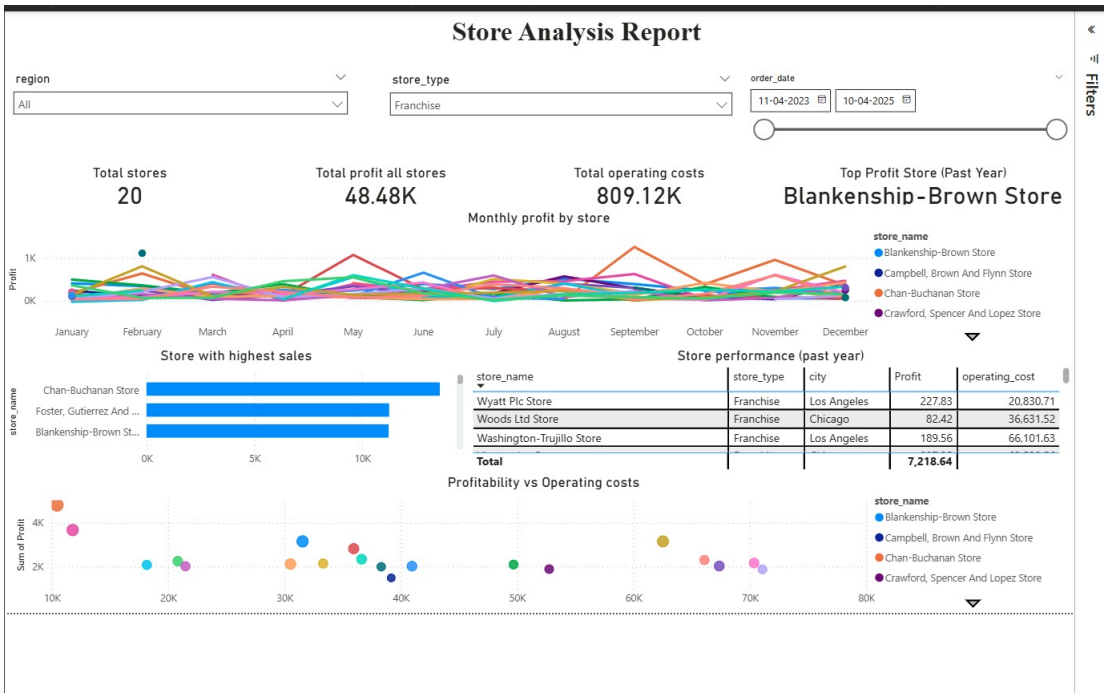

Figure 6: Customer Insights Report with age distribution, revenue by age group, regional customer distribution, and top customer profitability analysis

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6. Return Management & Quality Control

6.1 Return Pattern Analysis

Return analysis reveals controllable and uncontrollable factors:

- **Total Returns:** 200 units across analysis period
- **Defective Products:** 31.5% of returns (controllable)
- **Wrong Item Shipped:** 26.5% of returns (controllable)
- **Combined Controllable Returns:** 58% of total returns
- **Uncontrollable Returns:** Late delivery, customer preference changes (42% of returns)

6.2 Category-Specific Return Insights

Return rates vary significantly across product categories:

- **Personal Care:** 54 returns (highest absolute count)
- **Home & Kitchen:** 52 returns
- **Apparel:** 47 returns
- **Electronics:** 47 returns
- **Consistent Return Rates:** All categories show similar return rate percentages

6.3 Regional Return Distribution

- **West Region:** 57 returns (highest concentration)
- **South Region:** 33 returns
- **North Region:** 26 returns
- **East Region:** 21 returns

Strategic Implication: West region operational or quality issues warrant investigation and corrective action.

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7. Key Business Insights & Recommendations

7.1 Strengths

1. **Strong Revenue Base:** $818K revenue demonstrates solid market presence and customer demand
2. **Channel Diversification:** 70.7% in-store with 29.3% online provides multi-channel platform
3. **Customer Loyalty:** High-value and frequent customer segments indicate strong retention potential
4. **Product Portfolio:** Balanced category distribution reduces dependency on single product line
5. **Positive Trend:** Declining return rates (0.70 to 0.37) show quality improvement initiatives working

7.2 Opportunities

1. **Online Expansion:** Online channel at 29.3% presents significant growth opportunity matching in-store dominance
2. **Franchise Optimization:** Franchise stores underperforming vs. flagship format; evaluate format conversion strategy
3. **Age Group Targeting:** 26-35 segment shows growth potential with balanced customer base
4. **Return Reduction:** 58% controllable returns can be eliminated through quality assurance and fulfillment improvements
5. **Geographic Expansion:** Balanced regional distribution suggests untapped expansion potential

7.3 Challenges & Action Items

1. **April 2025 Decline:** 82% revenue drop from $42.5K to $8K requires emergency root cause analysis
    - Action: Implement forensic analysis of sales, market conditions, competitive activity in April 2025
2. **West Region Returns:** Elevated return concentration (57 of 200 units) indicates quality or operational issues
    - Action: Regional quality audit and operational process review
3. **Operating Cost Management:** $809.12K operating costs against $48.48K profit indicates margin pressure
    - Action: Implement cost optimization initiatives and operational efficiency improvements

4. **Customer Acquisition:** Relatively stable customer base suggests limited growth in new customer acquisition
    - Action: Launch targeted customer acquisition campaign focusing on underrepresented age segments

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8. Strategic Recommendations

8.1 Short-Term Actions (0-3 months)

1. **Emergency Response:** Investigate and address April 2025 revenue decline
2. **Quality Initiative:** Implement defect reduction program targeting 31.5% defective returns
3. **Fulfillment Accuracy:** Process review to reduce 26.5% wrong-item shipments
4. **West Region Audit:** Conduct operational and quality assessment in West region

8.2 Medium-Term Initiatives (3-9 months)

1. **Online Growth Strategy:** Develop digital transformation roadmap to grow online from 29.3% to 40%+ of revenue
2. **Customer Experience:** Implement loyalty programs targeting high-value (26-35 age group) and frequent purchaser segments
3. **Store Format Optimization:** Evaluate flagship vs. franchise performance and develop format strategy
4. **Product Innovation:** Leverage strong Personal Care category performance; identify expansion opportunities

8.3 Long-Term Strategic Goals (9-24 months)

1. **Omnichannel Excellence:** Integrate online and in-store channels for seamless customer experience
2. **Geographic Expansion:** Scale successful locations and formats across new markets
3. **Market Leadership:** Target market share growth through customer acquisition and retention
4. **Operational Excellence:** Achieve best-in-class operational metrics and profitability targets

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9. Conclusion

This comprehensive analysis reveals a retail organization with **strong fundamentals, clear growth potential, and specific operational challenges requiring immediate attention**. The $818K revenue base, diversified product portfolio, and multi-channel platform provide a solid foundation for growth.

Critical Success Factors:

1. Address April 2025 revenue decline root causes
2. Execute quality assurance program to reduce controllable returns (58% of total)
3. Accelerate online channel growth as primary expansion lever
4. Implement customer segmentation strategies for targeted retention and acquisition
5. Optimize store network performance through format and operational improvements

With disciplined execution of these recommendations, the organization can transform current challenges into competitive advantages, positioning for sustainable growth and market leadership in the retail sector.

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